Telecom Customer Segmentation

Business Insights Report

Project: ML-Based Customer Segmentation & Anomaly Detection
Dataset: IBM Telco Churn | Algorithm: KMeans ($\mathrm{k}=2$) + Isolation Forest | June 2025

Executive Summary

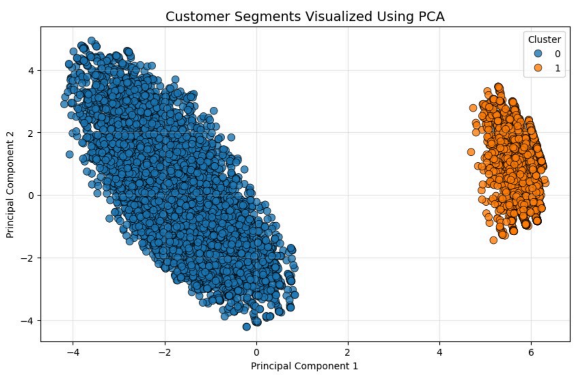
This report presents the findings of an end-to-end machine learning analysis conducted on a telecom customer dataset of 7,043 records. Using KMeans clustering ($\mathrm{k}=2$) and Isolation Forest anomaly detection, customers were segmented into two distinct behavioural groups. The analysis surfaces a critical retention risk: a high-value majority segment with a $\mathbf{31.8\%}$ churn rate, representing significant recurring revenue under threat. Three data-driven recommendations are presented to reduce churn, drive upsell, and prioritise at-risk account review.

Methodology

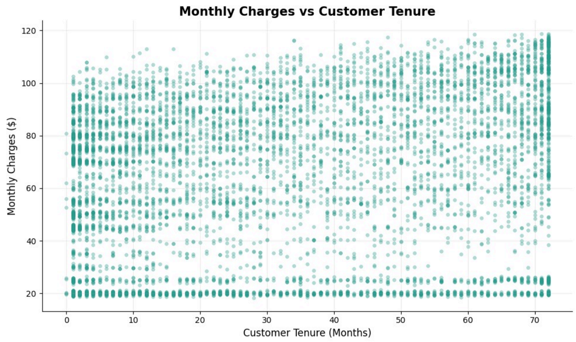
The analysis pipeline was built end-to-end in Python (Google Colab) and follows a structured, production-aligned workflow:

Stage	Tool / Technique	Purpose
1. Data Preparation	Pandas, NumPy	Label encoding, missing value imputation, type casting
2. Feature Scaling	StandardScaler	Normalise MonthlyCharges, TotalCharges, tenure for clustering
3. Optimal $\mathbf{k}$ Selection	Elbow Method + Silhouette Score	Determined $\mathrm{k}=2$ as optimal cluster count
4. Clustering	KMeans ($\mathrm{k}=2$, random_state=42)	Assigned each customer to a behavioural segment
5. Anomaly Detection	Isolation Forest (contamination=5%)	Flagged 353 statistical outliers across the dataset
6. Validation	Cluster profiling + business logic check	Verified segments align with domain-expected behaviour
7. Visualisation	Power BI + Custom HTML Dashboard	Interactive dashboard for stakeholder exploration

Customer Segments Visualized Using Principal Components Analysis (PCA)



Relationship Between Monthly Charges and Customer Tenure



Cluster Profiles - Comparative Analysis

KMeans identified two structurally distinct customer groups. The segmentation boundary is primarily driven by internet service adoption, which cascades into different pricing tiers, contract preferences, and churn behaviours:

Attribute	Cluster 0 - High-Value / High-Risk	Cluster 1 - Low-Value / Stable
Segment Size	5,517 customers (78.3%)	1,526 customers (21.7%)
Avg Monthly Charges	$76.84	$21.08
Avg Total Charges	$2,727	$663
Avg Tenure	32.9 months	30.6 months
Churn Rate	31.83%	7.40% ✓

Attribute	Cluster 0 - High-Value / High-Risk	Cluster 1 - Low-Value / Stable
Dominant Contract	60.7% Month-to-Month	41.8% Two-Year
Internet Service	56% Fiber / 44% DSL	100% No Internet (Phone Only)
Payment Method	Electronic Check (high risk)	Mixed - more auto-pay
Senior Citizens	Proportional representation	Low representation
Anomalies Flagged	~330 (majority share)	~23 (minimal)
Revenue Contribution	~84% of total revenue	~16% of total revenue
Risk Level	HIGH	LOW
Growth Opportunity	Retention / contract conversion	Internet upsell

Key Findings

- The churn gap is 4.3x: Cluster 0 churns at 31.83% versus 7.40% in Cluster 1. This is not a marginal difference - it indicates a structurally different relationship with the service, driven primarily by contract flexibility and higher spend expectations.
- Internet service is the true segmentation driver: Cluster 1 contains exclusively phone-only customers (no internet service). This means the segmentation boundary is not just a spend tier - it reflects a fundamentally different product bundle. Cluster 0 customers have more to lose if the service disappoints, and more alternatives to switch to.
- Month-to-Month contracts are the churn engine: 60.7% of Cluster 0 customers are on Month-to-Month contracts, which carry zero switching cost. In contrast, Cluster 1's 41.8% Two-Year contract rate explains their stability. Lock-in is the most direct lever available to reduce Cluster 0 churn.

- $\$134,936$ /month is the revenue at risk: Applying Cluster 0's churn rate to their average monthly spend gives a monthly revenue exposure of approximately $\$135\ \mathrm{K}$. On an annualised basis, this represents over $\$1.6\mathrm{M}$ in potential lost recurring revenue - a material business risk.
- 353 anomalies require individual review: Isolation Forest flagged 5% of the dataset as statistical outliers, concentrated in Cluster 0. These customers exhibit spending or usage patterns inconsistent with their segment peers - they may represent early churn signals, billing anomalies, or high-value accounts deserving proactive outreach.

Business Recommendations

1) Convert Cluster 0 to Long-Term Contracts (Priority: High)

The highest-impact lever available. With 60.7% of Cluster 0 on Month-to-Month contracts, a targeted incentive campaign - such as a 10-15% discount for switching to a One-Year or Two-Year plan directly removes the zero-switching-cost dynamic. Industry benchmarks suggest contract conversion can reduce monthly churn by $40-60\%$ in high-risk segments. Prioritise customers with tenure under 12 months and Electronic Check payment, as this sub-group shows the highest churn concentration.

2) Upsell Internet Services to Cluster 1 Phone-Only Customers (Priority: Medium)

Cluster 1 customers are stable, low-churn, and currently generating only $\$21$ /month on average versus $\$76/$ month for Cluster 0 . Since 100% of this segment has no internet service, a structured upsell campaign offering DSL or Fiber bundles represents the clearest growth path. Converting even 20% of Cluster 1 to a $\$50/$ month internet bundle adds approximately $\$3,052/$ month in incremental revenue with minimal churn risk, given this segment's demonstrated loyalty.

3) Proactive Outreach for 353 Anomaly-Flagged Accounts (Priority: Medium)

The Isolation Forest model flagged 353 customers whose behaviour deviates statistically from their cluster peers. These accounts warrant manual review: some may be billing errors, others may be high-value customers showing early churn signals (sudden drop in service usage, payment method changes). Assigning these accounts to a dedicated retention team or automated re-engagement workflow can recover value before it is lost. A 30-day review cycle is recommended.

Anomaly Detection - Isolation Forest Results

Isolation Forest was applied with a contamination parameter of 5%, meaning the model was calibrated to flag approximately 5% of the population as outliers. The results:

Metric	Value
Total Anomalies Flagged	353 customers
As % of Dataset	5.01%
Cluster Concentration	$\sim 93\%$ in Cluster 0 (High-Value / High-Risk)
Isolation Forest Label	-1 = Anomaly | +1 = Normal
Likely Profiles	Unusually high spend, extreme tenure, atypical payment patterns
Recommended Action	Manual review + proactive retention outreach within 30 days

Limitations & Assumptions:

The following constraints should be considered when acting on these findings:

- Dataset is cross-sectional - no time-series data available. Churn trends over time and seasonality effects cannot be assessed.

- $\mathrm{k}=2$ was selected via Elbow Method and Silhouette Score, but a $\mathrm{k}=3$ model (adding a mid-tier segment) may reveal additional nuance in Cluster 0.
- Revenue-at-risk calculations assume constant monthly charges and uniform churn probability within the cluster, which is a simplification.
- Anomaly detection results should be validated against CRM records before operational action - some flagged accounts may reflect legitimate edge cases (e.g., corporate accounts, trial users).
- No demographic data beyond gender and senior citizen flag is available; richer segmentation could incorporate usage frequency, support ticket history, or geographic data.

Conclusion

This analysis demonstrates that KMeans clustering applied to telecom customer data can surface actionable, high-stakes business intelligence beyond what standard reporting captures. The identification of a high-value majority segment with a 31.8% churn rate - representing over $\$1.6\mathrm{M}$ in annualised revenue risk - provides a clear, quantified mandate for targeted retention investment.

The three recommendations presented are deliberately sequenced by impact and feasibility: contract conversion addresses the root cause of Cluster 0 churn; internet upsell unlocks growth from Cluster 1’s stable base; and anomaly-flagged account review protects against silent attrition in the highest-spend customers. Together, they translate a machine learning output into a prioritised, revenue-linked action plan.

The accompanying interactive dashboard enables stakeholders to explore these findings at their own depth from executive-level KPIs to individual customer-level filtering - making this analysis both rigorous and accessible.